Faculty of Information Technology

Image Processing (in English) (ZPOe 25/26L)

Application of Basic Image Processing Filters on Human Face Images

1. Team Members

- Adar Otieno (xotiena00)
- Deniz Aksoy (xaksoyd00)
- Eyobed Awel (xnuriey00)

2. Project Title and Assignment
   Application and Evaluation of Basic Image Processing Filters on Human Face Images

Assignment Description

The objective of this project is to implement several basic image processing filters and analyze their visual effects when applied to human face images. Image filtering techniques are widely used in computer vision and digital image processing to enhance image quality, highlight important features, or create visually meaningful transformations.

In this project, three different image filters will be implemented using Python / C++ OpenCV and tested on a dataset of human face images. The selected filters represent different categories of image processing operations:

- Gaussian Blur – smoothing filter used to reduce noise and details
- Sharpen – enhancement filter used to emphasize image edges and details
- Emboss – artistic filter producing a relief-like effect

The goal of the project is to observe and compare how these filters affect visual characteristics of facial images.

3. Description of the Proposed Solution

The proposed solution consists of implementing several basic image processing filters and applying them to images from a human face dataset.

The workflow of the project will include the following steps:

1. Loading images from the dataset.
2. Implementing selected image processing filters .

3. Applying the filters to the dataset images.
4. Comparing the original and filtered images.
5. Evaluating the visual effects produced by each filter.

Each filter will be implemented convolution-based image processing techniques, which are commonly used in digital image processing.

The project will also include a simple demonstration program that allows users to apply the implemented filters to selected images from the dataset.

The objective is to demonstrate how relatively simple image processing operations can significantly change the appearance and characteristics of images.

4. Dataset Description

The project will use a Human Faces Dataset obtained from Kaggle.

The dataset contains a collection of images of human faces captured under different conditions such as:

- different lighting environments
- variations in facial expressions
- different backgrounds
- varying image resolutions

This type of dataset is particularly suitable for testing image processing filters because facial images contain important visual features such as edges, textures, and gradients.

These characteristics allow us to clearly observe the impact of various filters on important visual elements of the images.

The dataset will be used as the input for testing and evaluating the implemented filters.

5. Division of Labour

The work will be divided among the team members according to the filters they will implement.

Team / Member	Task
Adar	Implementation of the Gaussian Blur filter

Team / Member	Task
Deniz	Implementation of the Sharpen filter
Eyobed	Implementation of the Emboss filter

Each team member will:

- study the theoretical background of the assigned filter,
- implement the filter algorithm using Python/C++ OpenCV library,
- apply the filter to the dataset images,
- evaluate the visual effects produced by the filter.

The final application will combine the work of all team members into one program that allows the application of all implemented filters.

Individual Contributions

6. Gaussian Blur – Adar Otieno

Summary of Studied Knowledge, Existing Procedures and Methods

Gaussian Blur is a widely used smoothing technique in digital image processing. Its main purpose is to reduce noise and fine details in an image by applying a Gaussian-weighted averaging process to neighboring pixels.

One of the most common procedures for implementing Gaussian Blur is convolution with a Gaussian kernel. Convolution-based filtering is a fundamental technique in image processing where each pixel value is recalculated using the weighted values of surrounding pixels.

Existing implementations of Gaussian Blur are available in several computer vision libraries such as OpenCV, where functions like GaussianBlur() perform this operation efficiently. Gaussian smoothing is often used as a preprocessing step before other operations such as edge detection or feature extraction.

The Gaussian kernel used in convolution typically follows a normal distribution, giving higher weight to pixels near the center and lower weight to pixels farther away.

Planned Technologies and Methods

The filtering process will rely on convolution-based image processing methods, using a Gaussian kernel such as:

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} /16$$

OpenCV functions such as cv::GaussianBlur() (C++) or cv2.GaussianBlur() (Python) may be used to apply the filter.

Proposed Implementation

The filter will smooth facial textures and reduce fine image details. The results will be compared with the original images in order to evaluate how smoothing affects facial features and image noise.

Literature and Sources

- Gonzalez, R. C., Woods, R. E. – *Digital Image Processing*
- OpenCV Documentation
- Kaggle Dataset Documentation

7. Emboss Filter – Eyobed Awel

Summary of Studied Knowledge, Existing Procedures and Methods

The Emboss filter is an artistic image processing technique used to highlight edges and create a relief-like appearance in images. The method works by emphasizing the intensity differences between neighboring pixels.

Existing emboss filtering techniques typically rely on gradient-based convolution kernels. These kernels detect directional changes in pixel intensity and create an effect that makes the image appear engraved or embossed.

Emboss filters are commonly implemented using convolution operations available in many image processing libraries, including OpenCV. By shifting pixel intensity values and emphasizing gradients, the filter produces a pseudo-3D visual effect.

Emboss filters are frequently used in artistic image transformations and visual effect applications.

Planned Technologies and Methods

The implementation of the emboss will rely on convolution with an emboss kernel such as:

$$\begin{bmatrix} -2 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

OpenCV functions such as cv::filter2D() (C++) or cv2.filter2D() (Python) will be used to apply the kernel to the images.

Proposed Implementation

The filter will emphasize edges and contours of human faces, creating a relief-style visual effect.

The resulting images will be compared with the original images to evaluate how the emboss effect modifies the perception of facial structures.

Literature and Sources

- Gonzalez, R. C., Woods, R. E. – *Digital Image Processing*
- Computer Vision Learning Materials
- OpenCV Documentation

8. Sharpen Filter – Deniz Aksoy

Summary of Studied Knowledge, Existing Procedures and Methods

Sharpening is an image processing technique used to enhance edges and improve the clarity of images. The method increases the contrast between neighboring pixels, making image details more visible.

Existing sharpening methods often use high-pass filtering techniques, where high-frequency components of the image (such as edges) are amplified. One common procedure is convolution with a sharpening kernel.

In practical implementations, sharpening filters are available in computer vision libraries such as OpenCV. These filters can be implemented using convolution operations or specialized functions designed for edge enhancement.

Sharpening is frequently used in photography, computer vision systems, and image enhancement applications.

Planned Technologies and Methods

The sharpening effect will be created using a convolution kernel such as:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

OpenCV convolution functions such as cv::filter2D() (C++) or cv2.filter2D() (Python) will be used to apply the sharpening operation.

Proposed Implementation

The filter will increase the visibility of edges and textures in facial images.

The results will be analyzed by comparing the sharpened images with the original images to observe how edge enhancement affects facial details.

Literature and Sources

- Gonzalez, R. C., Woods, R. E. – *Digital Image Processing*
- OpenCV Documentation
- Kaggle Dataset Documentation

9. Conclusion

This project proposes the implementation and evaluation of three classical image processing filters applied to a human face dataset.

The selected filters represent different types of image transformations including smoothing, edge enhancement, and artistic effects.

By applying these filters to facial images, the project aims to demonstrate how basic image processing techniques can significantly alter visual characteristics of images.

The results will be evaluated by comparing original and filtered images from the dataset.